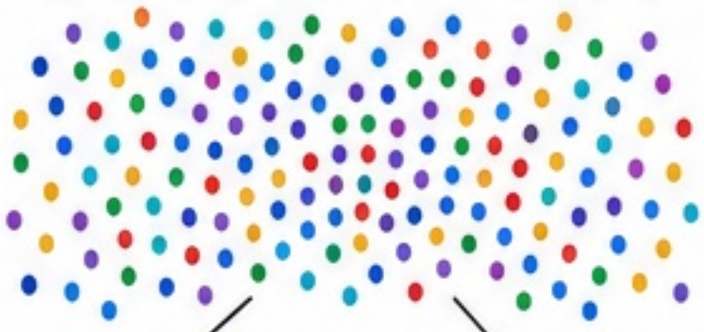
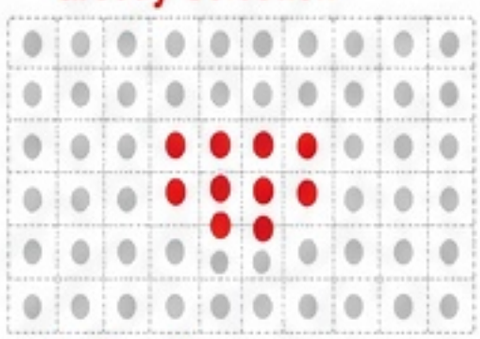
Synthesis Collapse: How Greedy Selection Narrows LLM Data Synthesis

The Problem – Synthesis Collapse

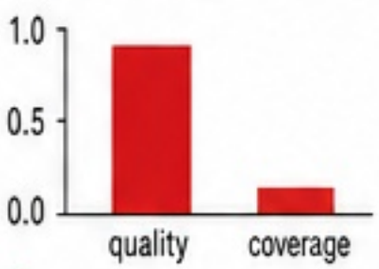
Synthetic sample pool (e.g., behavioral space)



Greedy Selection



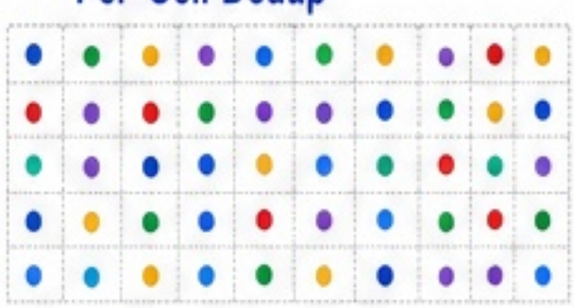
39 cells (80% waste)



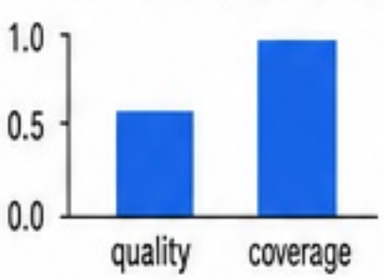
2.4× collapse

Conditionally invisible on benchmarks

Per-Cell Dedup



40 cells (100% fill)



Diagnose: when does coverage matter?

Predictive Framework

$$\eta^2 = \frac{\text{signal}}{\text{signal} + \text{noise}}$$

Detectable	Ceiling	Noise
$\eta^2 \gg 0$, coverage gap visible, 1.5B below ceiling	$\beta \approx 0$, model saturated, 7B 99–100%	$\sigma^2\varepsilon \gg$ signal, LoRA noise dominates, multi-seed

LoRA rank $r \uparrow$

P1 ↑ QD-Greedy gap grows monotonically

P2 ↓ Random-QD gap shrinks, noise artifact

Predict and Validate: 8/8 confirmed

Experimental Validation

Code 1.5B (8 seed)	Detectable (d = 8.58)
Code 7B (self-synth)	Ceiling (99–100%)
LoRA rank sweep	P1 + P2 confirmed
Noise injection	Invisible (∞× correctness)
Math cross-domain	No inverted-U predicted
Dialogue 8-seed	Invisible under LoRA noise

Per-cell dedup = mechanism (p = 0.98 vs full MAP-Elites)

Descriptor design = non-trivial component